

PORTFOLIO PROJECT 2

Customer Churn Analysis, RFM & Segmentation

Identifying high-value customers and churn risk through behavioral analysis

SQL · BigQuery

RFM Analysis

Dec 2010 – Dec 2011

Business Problem

What happens when businesses don't know who their customers really are?

The Challenge

Without a structured view of customer behavior, businesses apply **blanket marketing strategies** — spending budget on low-value customers while neglecting high-value ones.

Retaining an existing customer costs **5–7× less** than acquiring a new one — yet most businesses have no system to identify who is at risk of leaving.

01

Who are our most and least valuable customers — and why?

02

Which customers are at risk of churning, and how much revenue is at stake?

03

What is the right action for each customer segment?

Dataset & Methodology

METHODOLOGY

How was the data cleaned and how were customers scored?

541,909

Raw Rows

397,884

Valid Rows
(after cleaning)

4,338

Unique
Customers

1 Year

Dec 2010–
Dec 2011

Data Cleaning Steps

✗ NULL CustomerID	135,080 rows (24.9%)
✗ Cancelled transactions (prefix 'C')	9,288 rows
✗ Invalid Quantity (≤ 0)	10,624 rows
✗ Invalid UnitPrice (≤ 0)	2,517 rows
✓ Valid rows retained	397,884 (73.4%)

RFM Framework

R Recency

Days since last purchase
Lower = more recent = better

Score 5→1 via NTILE(5)

F Frequency

Number of unique invoices
Higher = more loyal = better

Score 1→5 via NTILE(5)

M Monetary

Total spend in £
Higher = more valuable = better

Score 1→5 via NTILE(5)

Churn threshold: 90 days of inactivity from snapshot date (2011-12-10)

Key Metrics

OVERVIEW

Business performance snapshot across 4,338 customers

£8.9M

Total
Revenue

18,532

Total
Orders

£2,054

Avg Customer
Lifetime Value

92.5

Avg Recency
(days)



38

countries
82% revenue from UK alone



4.3

avg orders
per customer



33.4%

churn rate
1,449 customers inactive 90+ days

RFM Customer Segmentation

8 behavioral groups — from Champions to Lost

 **Champions**

22.2%

962 customers

65.2% revenue

 **Loyal Customers**

23.0%

998 customers

16.5% revenue

 **At Risk**

10.5%

454 customers

8.3% revenue

 **Hibernating**

5.6%

241 customers

3.7% revenue

 **Potential**

7.4%

319 customers

1.6% revenue

 **Promising**

8.1%

351 customers

1.8% revenue

 **Cant Lose**

0.3%

13 customers

0.05% revenue

 **Lost**

23.1%

1,000 customers

2.7% revenue

 **Key Insight:** Top 2 segments (Champions + Loyal) = 45.2% of customers but generate 81.7% of total revenue

Champions — The Core Business Engine

FINDING 02

22.2% of customers. 65.2% of revenue. Non-negotiable to retain.

65.2%

of total revenue
from only 962 customers

£5,809,341

total segment revenue

0% churn rate



Avg Recency

13 days

Purchased just 2 weeks ago on average



Avg Frequency

11.1 orders

Buys consistently — deeply habitual



Avg Spend

£6,039

Almost 3× the overall customer average



Action: Exclusive loyalty program, early product access, dedicated account treatment

Churn Analysis

FINDING 03

33.4% of customers haven't purchased in 90+ days — £1M+ in revenue is silent

33.4%

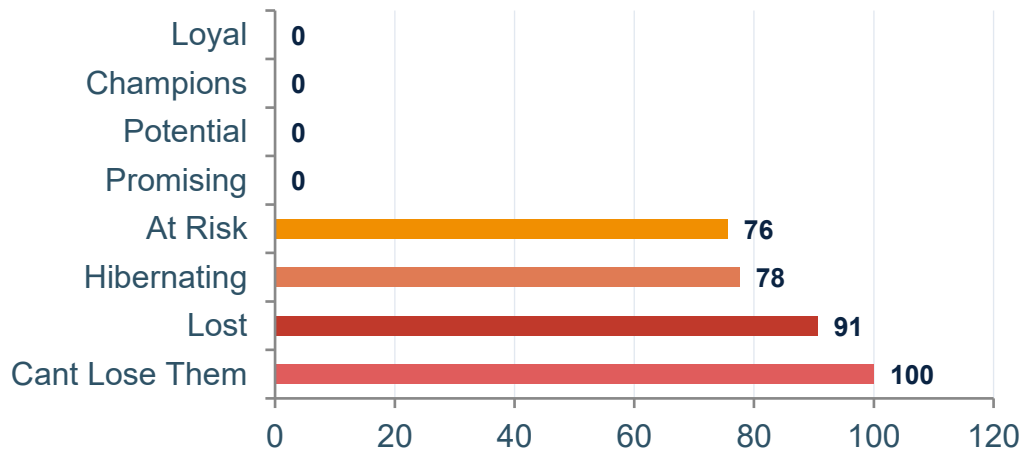
Overall
Churn Rate

1,449

Churned
Customers

£1,035,270

Revenue
at Risk



At Risk — 75.6% churn

£742,150 at risk

Highest-value win-back target

Cant Lose Them — 100% churn

£4,217 at risk

Personal outreach immediately

Hibernating — 77.6% churn

£329,638 at risk

Low-cost reactivation campaign

💡 At Risk is the priority: proven purchase history (avg 3.8 orders, £1,635 spend) but silent for 142 days — highest ROI win-back opportunity

At Risk — The Highest Priority Win-Back Segment

454 customers · 75.6% churn rate · £742,150 in revenue at stake

Why At Risk deserves the most urgent attention:

Metric	At Risk	Lost
Customers	454	1,000
Avg Recency	142 days	219 days
Avg Frequency	3.8 orders	1.3 orders
Avg Spend	£1,635	£244
Churn Rate	75.6%	90.6%
Total Revenue at Risk	£742,150	£243,935

Win-Back Strategy

1 Identify

Pull At Risk list from BigQuery — filter Recency 90–180 days + Monetary ≥ £1,000

2 Personalize

Craft email referencing their last purchase category — show you know them

3 Incentivize

Time-limited offer (10–15% off) — urgency drives response rates

4 Monitor

Track response at 2, 4, and 8 weeks post-campaign

Business Recommendations

RECOMMENDATIONS

Segment-specific actions ranked by priority and potential revenue impact

CRITICAL Champions

Launch exclusive loyalty program — early access, VIP perks, dedicated support

 Protects £5.8M (65.2% of revenue)

CRITICAL At Risk

Personalized win-back email + time-limited 10–15% discount within 2 weeks

 £742,150 revenue at risk — highest ROI intervention

HIGH Loyal Customers

Upsell campaigns, product recommendations, frequency incentives

 £1.47M segment — natural spend-increase candidates

HIGH Cant Lose Them

Direct personal outreach — call or personalized email immediately

 100% churned — high purchase history, worth saving

MEDIUM Potential Loyalists

Welcome series, product education, gentle nudge campaigns

 Recent buyers (19 days) not yet habitual — window open

LOW Lost

Exit survey only — do not invest in paid re-engagement

 2.7% revenue contribution — re-acquisition cost too high

Customer retention is not a marketing tactic — it is a revenue protection strategy.

Thank You

Customer Churn Analysis, RFM & Segmentation · Portfolio Project 2

4,338

Customers
Analyzed

8 Segments

RFM-Based
Behavioral Groups

£1,035,270

Revenue at Risk
Quantified

Connect & Explore



GitHub

github.com/retatriasriyani



Dashboard

Looker Studio — Interactive Dashboard (link to be added)



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